

Testing the prototype with its users

The plan, the information sheet and consent form, the task sheets for patients, clinicians, pathologists and administrators, the System Usability Scale and the results template, for the user acceptance test of the research proposal (§3.8.2).

AI-Driven Mobile Health Platform for Prostate Cancer Diagnosis (Zambia)

Made on 1 October 2026

For use only after ethics approval. Participants use synthetic data only.

User acceptance testing (UAT): the plan

This kit is for the user acceptance test that the research proposal describes (§3.8.2, NFR-04, NFR-11). It turns that design into sessions you can run: who takes part, what they do, what you record, and how you analyse it. The printable version of every document is [docs/uat/PCa-mHealth-UAT-kit.pdf](#).

Document	What it is for
This plan	Aims, participants, ethics, preparation, how a session runs, how to analyse the results
Information sheet and consent form	Given to every participant before the session
Tasks: patients	Tasks, what success looks like, and the observer's record
Tasks: clinicians	For urologists, general practitioners, nurses and radiographers (the app's clinician role)
Tasks: pathologists	
Tasks: administrators	For the admin website (optional: the proposal's UAT does not include administrators)
System Usability Scale (SUS)	The ten statements, how to read them aloud, and how to score them
Results template	Tables for the report's Chapter 4 (test output)

1. Aims

1. Can each kind of user complete the main tasks of their role without help? (task success, time, errors, help needed)
2. Is the system easy to use? The proposal's target is a **SUS score of at least 75** (NFR-04).
3. Can people with limited reading skills use the assistant and the Learn section by **voice and read-aloud?** (think-aloud, proposal §3.8.2)
4. What should change before the next prototype iteration? (problems ranked by severity)

2. Before any participant: ethics

No session with real participants may start before written approval from the ZCAS University Ethics Review Board and, for health-facility staff and patients, the Zambia National Health Research Authority (NHRA). This is what the proposal commits to (§3.7.1).

- **Only synthetic data.** Participants use test accounts and made-up patients prepared by you. Nobody enters their own or a real patient's details. Tell them so before they start, and stop them politely if they begin to.

- **Consent first.** Every participant reads (or is read) the [information sheet](#) and signs or thumb-prints the consent form, with a witness for those who cannot read.
- **What you record about participants:** a code (for example P-03, C-07), their role, years in the role, how often they use a smartphone, and the language they prefer. No names on the record sheets; the list linking codes to names is kept separately and locked.
- **Audio recording** of think-aloud sessions only if the participant ticks that box on the consent form. Otherwise, take notes.
- **Patients** are told that the app gives general information only, that the AI results in the prototype are not real, and that nothing they see is about their own health.

3. Participants

The proposal asks for **at least five people in each user group**, from Cancer Diseases Hospital (CDH), University Teaching Hospital (UTH), and at least one rural or peri-urban facility.

Group in the proposal	Account they use in the app	Task sheet
Urologists, general practitioners, nursing staff	Clinician	tasks-clinician.md
Radiologists (or radiographers)	Clinician (uploading scans, reading the AI report)	tasks-clinician.md , tasks C-6 to C-8
Pathologists	Pathologist	tasks-pathologist.md
Patients (men aged 40 and over, including some with limited reading skills)	Patient	tasks-patient.md
Administrators (optional)	Admin website	tasks-administrator.md

Five users per group find most usability problems of a kind (Nielsen and Landauer, 1993); more are better for the SUS average.

4. Preparation (the day before)

- ☐ Everything starts: `powershell -ExecutionPolicy Bypass -File 6-infrastructure\scripts\dev-up.ps1` ends with READY.
- ☐ The demo data exists and passwords are known (`SEED_DEMO_PASSWORD` in `.env` ; see the operations manual §14).
- ☐ **One fresh account per participant**, made on the admin website (Users > New account), named with the participant's code, for example *P-03 Test*. Patients' accounts are linked to a synthetic patient record (Patient accounts > Link).
- ☐ The test files are on the phone: `npm run fixtures` in `3-application-logic\backend` , then copy `synthetic-mri.dcm` and `synthetic-slide.tif` from `test\fixtures\files` to the phone's Downloads.
- ☐ The phone (or phones) charged, the app installed (operations manual §10.2) and the **volume up** for read-aloud. Leave the microphone permission unanswered if you want to see how participants handle Android's question; otherwise allow it beforehand.

- ☐ For the "no internet" tasks: you know how to switch **Airplane mode** on and off on that phone.
- ☐ Printed: the information sheet, consent forms, the task sheet for each participant, the SUS form, a stopwatch or phone timer.

5. A session (about 45 minutes, one participant)

Step	Minutes	What happens
Welcome and consent	5	Explain the aim (testing the system, not the person). Information sheet, consent form.
Background questions	3	Role, years, smartphone use, preferred language (on the task sheet).
Practice	2	One easy task (sign in), so the think-aloud feels normal.
Tasks	20–25	The participant does the tasks on their sheet, thinking aloud. The facilitator reads each task as written and does not help unless asked twice. The observer times each task and records the result.
SUS	5	The ten statements, read aloud if the participant prefers.
Interview	5–10	The questions at the end of the task sheet.
Close	1	Thank them; reset the phone (sign out, switch Airplane mode off).

Think-aloud: ask the participant to say what they are looking for, what they expect to happen, and what surprises them. If they fall silent for a while, say only "*What are you thinking now?*" Do not explain the screen.

Help levels (record one per task): **0** no help; **1** a hint ("look at the top of the screen"); **2** the facilitator showed them; **X** gave up.

Success: the task's success criterion on the sheet is met with help level 0 or 1.

6. Analysis

1. **Task success rate** per task and group: tasks completed with help 0 or 1 ÷ participants.
2. **Time on task:** median and range per task (only successful attempts).
3. **SUS:** score each form as in [sus-questionnaire.md](#); report the mean, the standard deviation and the range per group and overall, and compare with the target of 75.
4. **Problems found:** list every problem seen or reported, how many participants met it, and its severity (Nielsen's scale: 0 not a problem, 1 cosmetic, 2 minor, 3 major, 4 catastrophic). Fix 3s and 4s before the next iteration.
5. **Think-aloud and interviews:** thematic analysis. Read the notes, give each remark a short code (for example *trust in the AI label*, *voice not understood*, *offline confusion*), group the codes into themes, and count how many participants mention each theme. Quote a few remarks (by code, never by name).

6. Put the results in the [results template](#) and in Chapter 4 of the report. Report what was found, including tasks that failed: they are the point of the test.

7. What this test cannot show

- The AI results are **labelled mock output**, so the test can show whether people understand and trust the report's layout and labels, not whether the AI is right.
- The knowledge base is a **draft** that a clinician has not yet signed off; note any answer a participant finds wrong or unclear, and give the list to the reviewing clinician.
- The app is **in English**. Patients who prefer Bemba or Nyanja can be helped by an interpreter, and the session notes should say so; this is evidence for the translation work the proposal plans.

Participant information sheet and consent form

*Template. Fill in the parts in [square brackets], and use the version approved by the ethics committees.
Translations into Bemba or Nyanja must be done and checked by a qualified translator before use.*

Information sheet

Study title: Testing a mobile health app for prostate cancer screening in Zambia (a research prototype)

Researcher: [Your name], final-year student, Bachelor of Science in Computer Science, ZCAS University.
Supervisor: [supervisor's name]. Contact: [phone or e-mail].

Ethics approval: ZCAS University Ethics Review Board, reference [number]; National Health Research Authority, reference [number].

What is this study about? I have built a test version of a phone app and a website. They help clinics record prostate cancer screening, help pathologists review slides, and give men general information about the prostate, including an assistant that answers questions and can read text aloud. I want to find out whether the app is easy to use, and what needs to change.

What will I be asked to do? You will use the app on a phone we give you, for about 45 minutes. You will be asked to do some everyday tasks, such as reading an article or recording a made-up screening, and to say aloud what you are thinking while you do them. Then you answer ten short questions about how easy it was, and a few questions about your experience.

Important: - Everything in the app is **made up (synthetic)**. Please do not type your own name, phone number, NRC number or any real patient's details. - The app is a **test version**. Its "AI results" are practice output, not real results, and nothing in it is about your health. - The assistant gives **general information only**. It is not a doctor. If you have a health concern, please speak to a clinician.

Do I have to take part? No. Taking part is your choice. You can stop at any time without giving a reason, and it will not affect your care or your work in any way.

What will happen to the information? We write notes about how the tasks went and your answers to the questions, under a code, not your name. The list that links your name to your code is kept locked and separately, and is destroyed when the study ends [date]. If you agree, we may record your voice while you think aloud; the recording is used only to check our notes and is deleted at the end of the study. Results are reported without names, and quotes are given under codes only.

Are there risks or benefits? There are no known risks beyond the time you give. There is no payment [or: you will receive ... for your time and transport]. Your feedback will help make the app easier to use.

Questions or complaints: the researcher [contact], the supervisor [contact], or the ethics committee [contact].

Consent form

Participant code: __

Please tick each box:

- ☐ I have read the information sheet, or it has been read to me, and I understand it.
- ☐ I have been able to ask questions, and they were answered.
- ☐ I understand that taking part is voluntary and that I can stop at any time without giving a reason.
- ☐ I understand that I must not enter real personal or health information, and that the app shows made-up data only.
- ☐ I agree that my answers and comments may be used in the research report, without my name.
- ☐ I agree that my voice may be recorded while I think aloud (optional; leave unticked to say no).

I agree to take part in this study.

	Name	Signature or thumbprint	Date
Participant			
Witness (if the participant cannot read)			
Researcher			

Tasks: patients

Before the session: a fresh patient account for this participant (for task P-1 they create one themselves; for the others, use the prepared account, linked to a synthetic patient with at least one screening record, one consent and one message). The phone signed out, volume up, internet on.

Participant code: __ Date: __ Facilitator: __ Observer: __

Background: age group (40–49 / 50–64 / 65+): _ • **Uses a smartphone:** *daily / sometimes / rarely* • **Reads English:** *easily / with difficulty / not at all* • **Preferred language:** ____

Read each task aloud **exactly as written**. Record the time from the end of reading to success (or giving up), the help level (**0** none, **1** a hint, **2** shown, **X** gave up), and what happened.

#	Task (read aloud)	Success when	Time	Help	Notes (errors, words said while thinking aloud)
P-1	"You are new. Make yourself an account in the app. Use the details on this card." (give a card with made-up details: a name such as <i>Test Patient 03</i> , an e-mail such as <i>p03@example.test</i> , a phone number such as 0977 000 003, an NRC such as 123456/78/1, and a password)	The screen says <i>Your account is ready</i>			
P-2	"Now sign in with the test account on this card." (prepared account; the app asks for a new password: give one on the card)	The patient home screen shows			
P-3	"Find out when your last screening was."	Says the date shown on Home			
P-4	"Find your last PSA result."	Opens Results and reads the value aloud (not asked what it means)			
P-5	"Find the article that explains what a PSA test is, and read the start of it."	The article <i>What is a PSA test?</i> is open			
P-6	"Now have the phone read that article to you. Make it read more slowly. Then stop it."	Listen plays; <i>Slower voice</i> chosen; reading stopped			
P-7	"Ask the assistant: what happens during a DRE?" (by typing)	The answer is shown			
P-8	"Where does that answer come from?"	Points to the source under the answer			

#	Task (read aloud)	Success when	Time	Help	Notes (errors, words said while thinking aloud)
P-9	"Ask it, by speaking: does it hurt?" (real phone only; skip on the emulator)	The spoken words appear and are sent; the answer is about the DRE			
P-10	"Have the phone read the answer to you."	<i>Listen</i> under the answer plays it			
P-11	"Ask the assistant something a friend might say, like hello or how are you."	A friendly reply is shown			
P-12	"Start a new conversation. Then go back to the one about the DRE."	The earlier chat reopens with its messages			
P-13	"You have a new message from your clinic. Read it."	The message is opened (it is marked read)			
P-14	"You no longer want your data used for research. Find where you would say so, but do not confirm it."	<i>Profile > My consents</i> is open and the <i>Withdraw</i> question is shown; then <i>Cancel</i>			

Think-aloud on the assistant (P-7 to P-12): note whether the participant trusts the answer, understands that it is general information, notices the disclaimer, and understands the source. For participants with limited reading skills, note whether *Listen* and voice let them do the task without reading.

Interview (after the SUS)

1. What did you like most? What was hardest?
2. Was anything confusing or worrying? Where?
3. The assistant: would you ask it questions about your health? Why, or why not? Did you trust its answers?
4. Reading aloud and speaking to the phone: were they useful to you? Was the voice clear?
5. In which language would you most like to use the app?
6. Would you recommend it to a friend or relative who is going for screening?

Tasks: clinicians (urologists, general practitioners, nurses, radiographers)

Before the session: a fresh clinician account for this participant at a synthetic facility, already signed in once (temporary password changed). `synthetic-mri.dcm` in the phone's Downloads. The phone online.

Participant code: __ Date: __ Facilitator: __ Observer: __

Background: role: urologist / GP / nurse / radiographer / other __ · **Years in the role:** __ · Uses a smartphone: daily / sometimes / rarely · Has used an electronic record system (for example SmartCare): yes / no

Read each task aloud **exactly as written**. Record the time, the help level (**0** none, **1** a hint, **2** shown, **X** gave up) and what happened. All names and numbers are made up: give the participant a card with them.

#	Task (read aloud)	Success when	Time	Help	Notes
C-1	"Sign in with the account on this card."	The clinician home is shown			
C-2	(Facilitator switches on Airplane mode .) "The clinic has no internet today. Register this new patient." (card: made-up name, date of birth, MRN, district)	The patient is listed with <i>Saved on device</i>			
C-3	"Record a screening for this patient: PSA 6.2, free PSA 1.1, DRE enlarged and smooth."	The record is listed with <i>Saved on device</i>			
C-4	Time pressure: "A queue is waiting. Register one more patient and record their PSA of 4.8 and a normal DRE, as quickly as you can." (record the time; the target is under 3 minutes)	Patient and record saved			
C-5	(Facilitator switches Airplane mode off .) "The internet is back. Make sure today's work reaches the server."	The badges show <i>Synced</i> (by waiting or <i>Sync now</i>)			
C-6	"The patient agrees to an AI analysis of their data. Record their consent."	<i>AI analysis</i> consent is listed as given			
C-7	(Airplane mode on again.) "Add the MRI file in Downloads to this patient's images." Then (Airplane mode off) "Make sure it reaches the server."	The file moves from <i>Waiting on this phone</i> to the server list			
C-8	"Ask for an AI analysis for this patient, and open the result when it is ready."	The report is open			
C-9	"Tell me what this report says, and how far you would rely on it."	Mentions the <i>development mock data</i> banner (records how they read the report)			
C-10	"Ask the assistant what a PI-RADS score of 4 means."	The answer with its source is shown			

#	Task (read aloud)	Success when	Time	Help	Notes
C-11	"Find all the AI results in your clinic."	<i>AI results to review</i> list is open			

Observe: whether the offline state (C-2 to C-5) is understood without explanation; whether the *mock* banner on the report is noticed unprompted (C-9); errors in data entry (wrong field, units).

Structured questions on the report (C-9): on a scale of 1 to 5, how clear is it (a) which parts of the AI ran, (b) why there are no explanation images, (c) that the result is not a clinical result? Note: the prototype's AI uses practice (mock) models, so this checks the report's design and labels, not the AI's accuracy.

Interview (after the SUS)

1. Which tasks would you do every day? Which were slow or awkward?
2. Working without internet: was it clear what was saved and what was sent?
3. The AI report: what would you need to see before you would use such a report in practice?
4. Is anything missing that you would need in your clinic?
5. Would the assistant be useful for your patients? For you?

Tasks: pathologists

Before the session: a fresh pathologist account, signed in once. A synced synthetic patient in the same facility, with an AI report already made (request one with a clinician account beforehand). `synthetic-slide.tif` in the phone's Downloads.

Participant code: __ Date: __ Facilitator: __ Observer: __

Background: years as a pathologist: ____ · Reviews prostate biopsies: often / sometimes / rarely · Uses a smartphone: daily / sometimes / rarely · Has used digital pathology: yes / no

#	Task (read aloud)	Success when	Time	Help	Notes
H-1	"Sign in with the account on this card."	The pathologist home is shown			
H-2	"Add the slide in Downloads to this patient's record. It is a TIFF file."	The slide is listed as <i>Waiting for a pathologist</i>			
H-3	"Find the slides waiting for your review."	The review queue is open			
H-4	"Review this slide: the primary pattern is 4 and the secondary pattern is 3. Save the review."	<i>Reviewed: Gleason 4 + 3 = 7, ISUP grade group 3</i>			
H-5	"What grade group did the system give? Do you agree with it?"	Reads the grade group			
H-6	"Open the AI analysis for this patient and tell me what it shows."	The report is open; mentions the mock banner			

Observe: whether the Gleason entry matches how pathologists think about patterns; whether "a slide can only be reviewed once" causes concern (note it, do not explain it first).

Interview (after the SUS)

1. Is recording the Gleason score this way practical? What would you change?
2. Would you review slides on a phone, a tablet or a computer?
3. What would an AI grading tool need to show before you would trust it as a second opinion?

Tasks: administrators (admin website)

Before the session: a fresh administrator account, signed in once on a computer with the admin website open (<http://localhost:5173>). A patient account made through the app (task P-1) that is not yet linked, and a synthetic patient record with the same NRC.

Participant code: __ Date: __ Facilitator: __ Observer: __

Background: role: __ · Years managing user accounts or IT systems: ____

#	Task (read aloud)	Success when	Time	Help	Notes
A-1	"A new clinician starts today at the Lusaka facility. Give them an account." (card: made-up name and e-mail)	The account is listed, with a one-time password shown			
A-2	"This clinician has forgotten their password. Give them a new one."	A new one-time password is shown			
A-3	"A patient made an account in the app. Connect it to their clinic record."	<i>Linked</i> is shown			
A-4	"A test account is no longer needed. Remove it."	<i>The account ... was deleted</i> (or the reason it cannot be, for an account with clinical history)			
A-5	"Check that nobody has changed or removed entries in the audit log."	The integrity check says the chain is intact			
A-6	"Prepare the data to send to SmartCare Pro for this facility, and download it."	The FHIR bundle is downloaded			
A-7	"How many people used the phone app this week?"	Reads it from the dashboard			

Interview (after the SUS)

1. Which tasks were clear, and which were not?
2. Is anything missing that an administrator at a hospital would need?
3. The audit log and the integrity check: would they help you answer questions from auditors?

System Usability Scale (SUS)

The System Usability Scale (Brooke, 1996) is ten statements answered from 1 (strongly disagree) to 5 (strongly agree). It is free to use; cite Brooke, J. (1996) "SUS: a 'quick and dirty' usability scale", in *Usability Evaluation in Industry*, Taylor & Francis.

Giving it: after the tasks and before the interview. Ask the participant to answer quickly, with their first impression, and to answer every statement (if unsure, the middle, 3). For participants who prefer it, **read each statement aloud** and point to the five boxes, or show five faces from very unhappy to very happy; write down the number they choose. Say "the system" means the app (or the website) they just used. Do not change the wording; translations must be checked by a qualified translator before use.

Participant code: __ Group: __ Date: __

#	Statement	Strongly disagree 1	2	3	4	Strongly agree 5
1	I think that I would like to use this system frequently.	☐	☐	☐	☐	☐
2	I found the system unnecessarily complex.	☐	☐	☐	☐	☐
3	I thought the system was easy to use.	☐	☐	☐	☐	☐
4	I think that I would need the support of a technical person to be able to use this system.	☐	☐	☐	☐	☐
5	I found the various functions in this system were well integrated.	☐	☐	☐	☐	☐
6	I thought there was too much inconsistency in this system.	☐	☐	☐	☐	☐
7	I would imagine that most people would learn to use this system very quickly.	☐	☐	☐	☐	☐
8	I found the system very cumbersome to use.	☐	☐	☐	☐	☐
9	I felt very confident using the system.	☐	☐	☐	☐	☐
10	I needed to learn a lot of things before I could get going with this system.	☐	☐	☐	☐	☐

Scoring

1. For statements **1, 3, 5, 7 and 9** (the positive ones): score = answer – 1.
2. For statements **2, 4, 6, 8 and 10** (the negative ones): score = 5 – answer.

3. Add the ten scores (0 to 40) and **multiply by 2.5**. The result is between 0 and 100. It is not a percentage.

Example: answers 4, 2, 4, 1, 4, 2, 5, 1, 4, 2 → odd statements $3 + 3 + 3 + 4 + 3 = 16$; even statements $3 + 4 + 3 + 4 + 3 = 17$; $(16 + 17) \times 2.5 = \mathbf{82.5}$.

Reading the score: the average across many published studies is about 68. The proposal's target is **at least 75** (NFR-04). Report the mean, standard deviation and range per user group and overall, and say how many people answered each group. A form with an unanswered statement is scored with 3 for that statement and the fact is noted.

UAT results (template for Chapter 4)

Fill this in from the task sheets and SUS forms. Leave a cell empty rather than guess, and report failures as they happened.

Sessions: dates __ · facilities __ · number of participants per group: patients __ · clinicians __ · pathologists __ · administrators __

1. Task success and time

Task	Participants	Succeeded (help 0 or 1)	Success rate	Median time (s)	Range (s)	Main problems seen
P-1 Create an account						
P-2 Sign in						
...						
C-4 Registration under time pressure (target < 3 min)						
...						

2. System Usability Scale

Group	n	Mean SUS	Standard deviation	Lowest	Highest	Target (≥ 75) met?
Patients						
Clinicians						
Pathologists						
Administrators						
All						

3. Problems found

#	Problem	Where (task, screen)	Participants who met it	Severity (0–4)	Proposed change
1					

Severity: 0 not a problem · 1 cosmetic · 2 minor · 3 major (fix before the next iteration) · 4 catastrophic (fix before any further use).

4. Themes from think-aloud and interviews

Theme	Codes grouped in it	Participants mentioning it	Example remark (by participant code)

5. Read-aloud and voice (participants with limited reading skills)

Participant	Reads English	P-6 read-aloud	P-9 voice	P-10 listen	Notes

6. Conclusion for the report

- Was the SUS target of 75 met, overall and per group?
- Which tasks failed, and why?
- What changes go into the next prototype iteration (severity 3 and 4 first)?